

CryptoSilicon



In a world where software can be hacked and keys can be leaked, how do we secure the very heart of our technology—the hardware itself?

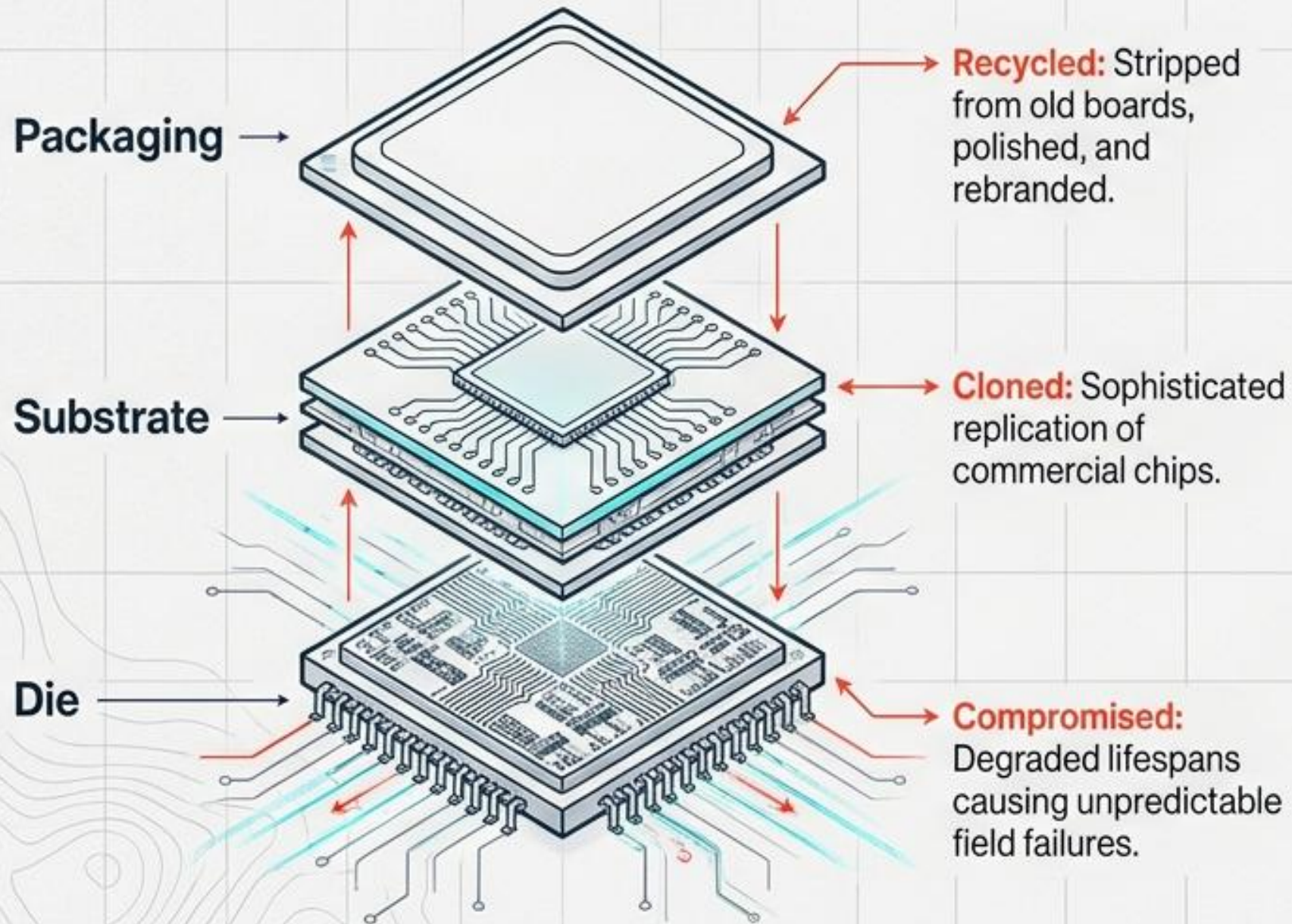
Project: Thermal-Transient Hardware Fingerprinting (PUF)

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Forum: Orbeetal Idea Contest 2026

The Multi-Billion Dollar Influx of Counterfeit Silicon

The Threat



The Impact



Billions of dollars lost annually across supply chains.



Catastrophic risks to avionics, defense grids, and medical devices.



Structural integrity compromised in secure cloud logistics.

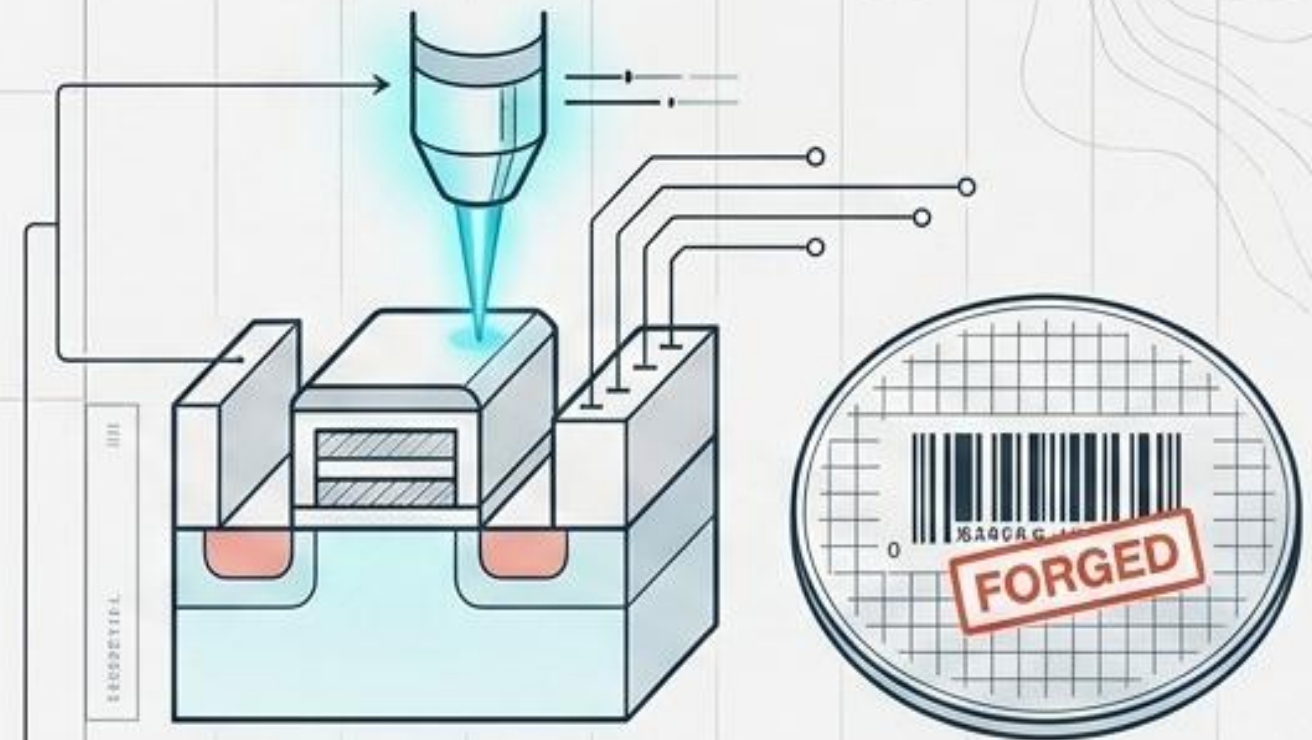
The Fundamental Flaw of Legacy Hardware Security

Software & Firmware Keys



Firmware encryptions are easily read and intercepted. If the code leaks, security drops to zero.

Non-Volatile Memory (NVM) & Labels

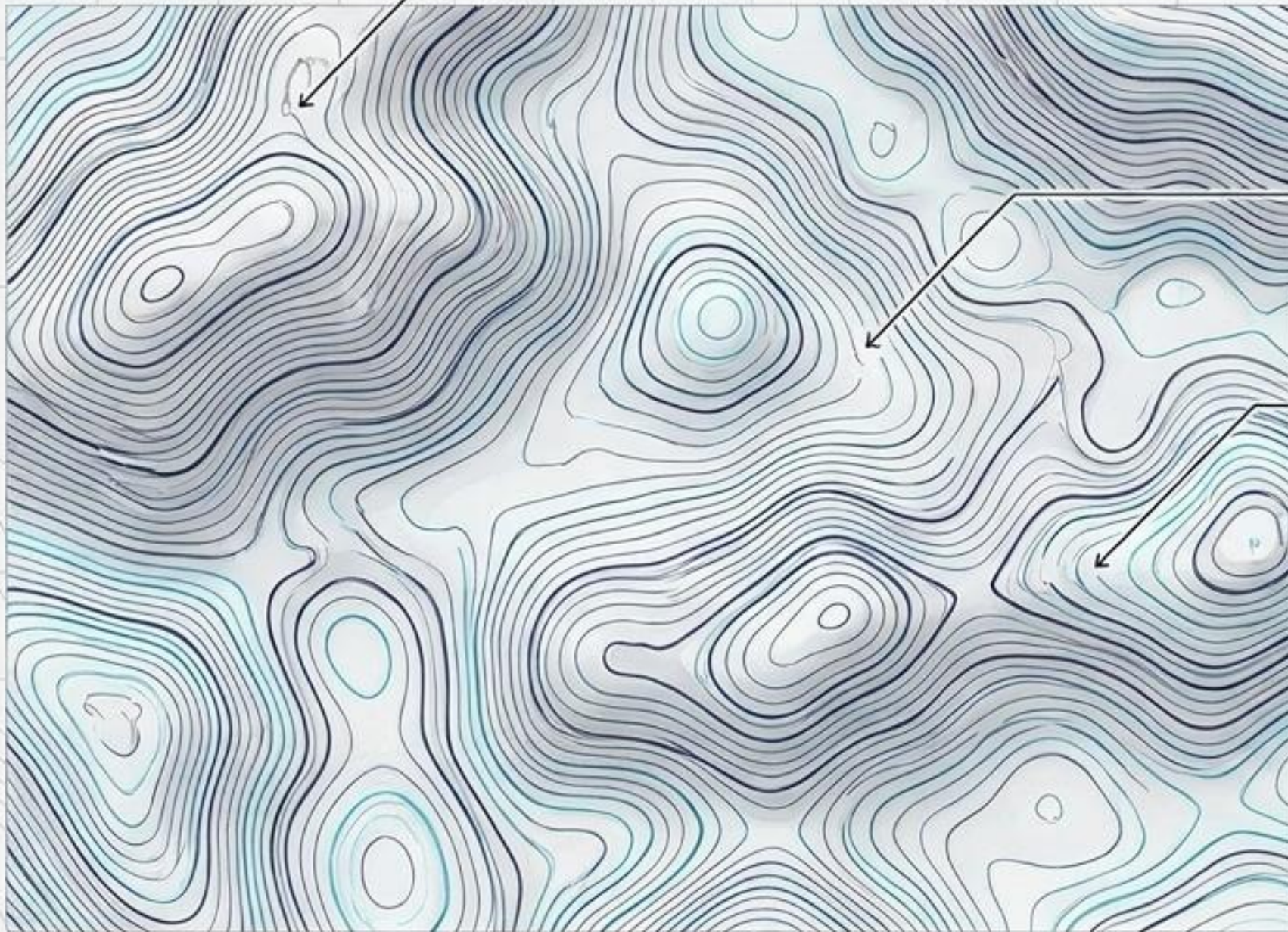


Permanent keys in NVM can be reverse-engineered or read using Focused Ion Beam (FIB) micro-probing. Surface barcodes are effortlessly forged.

Current methods secure the state of the memory, not the physics of the device.

Exploiting Atomic Randomness as a Cryptographic Asset

Microscopic Topography



Photolithography Anomalies:
Deterministic and stochastic variations during fabrication.

Chemical-Mechanical Planarization (CMP):
Microscopic surface inconsistencies.

Ion Implantation Variances:
Random dopant concentration profiles and line-edge roughness.

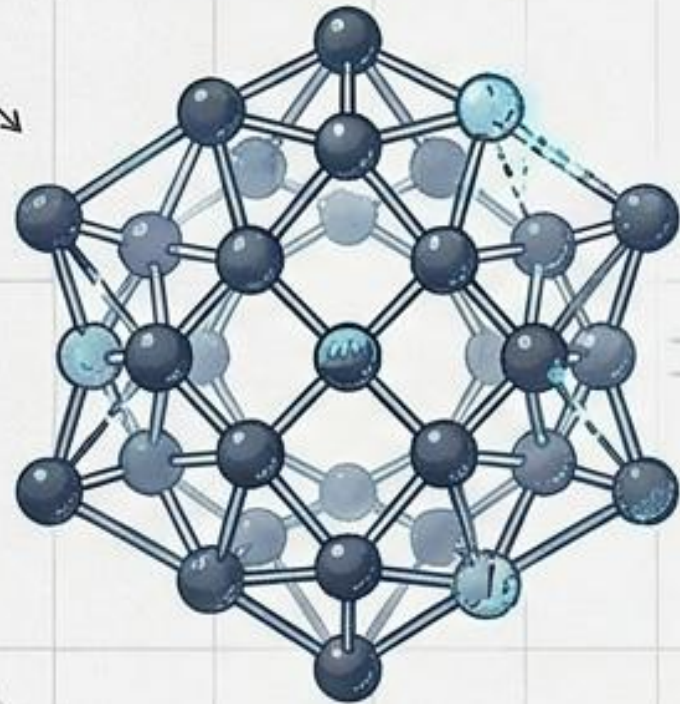
Key Takeaway

Physical Unclonable Functions (PUF)

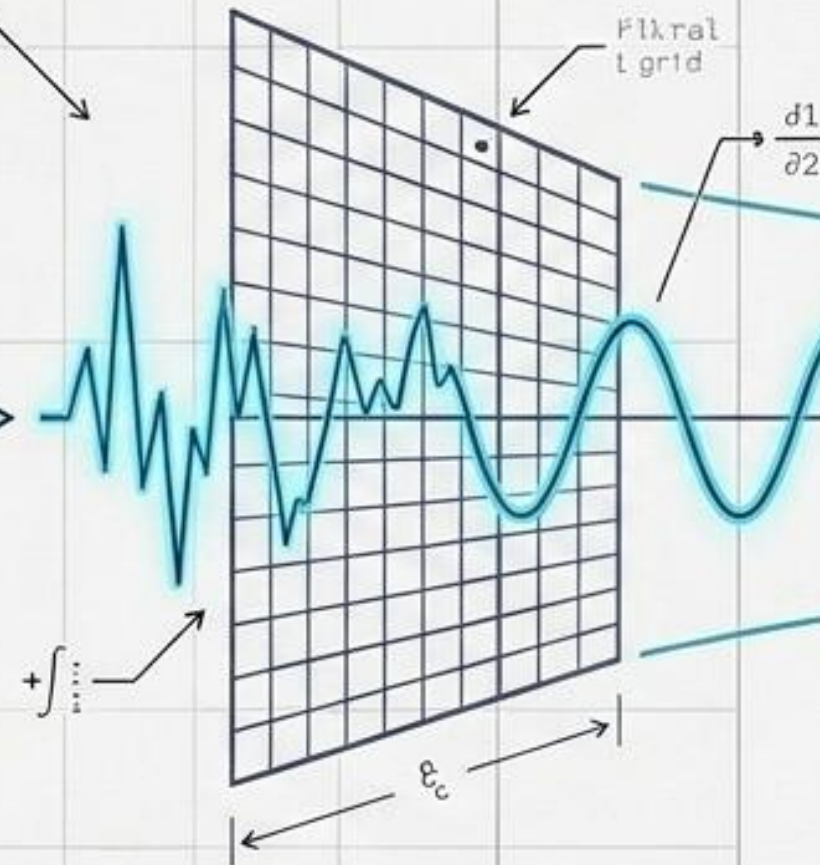
No two microchips can ever be fabricated with identical molecular structures. We are shifting security from vulnerable digital registers directly into structural device physics.

The Alchemy of CryptoSilicon

Raw Physical Variance
(Entropy)



Thermal Transient + Wavelet Math

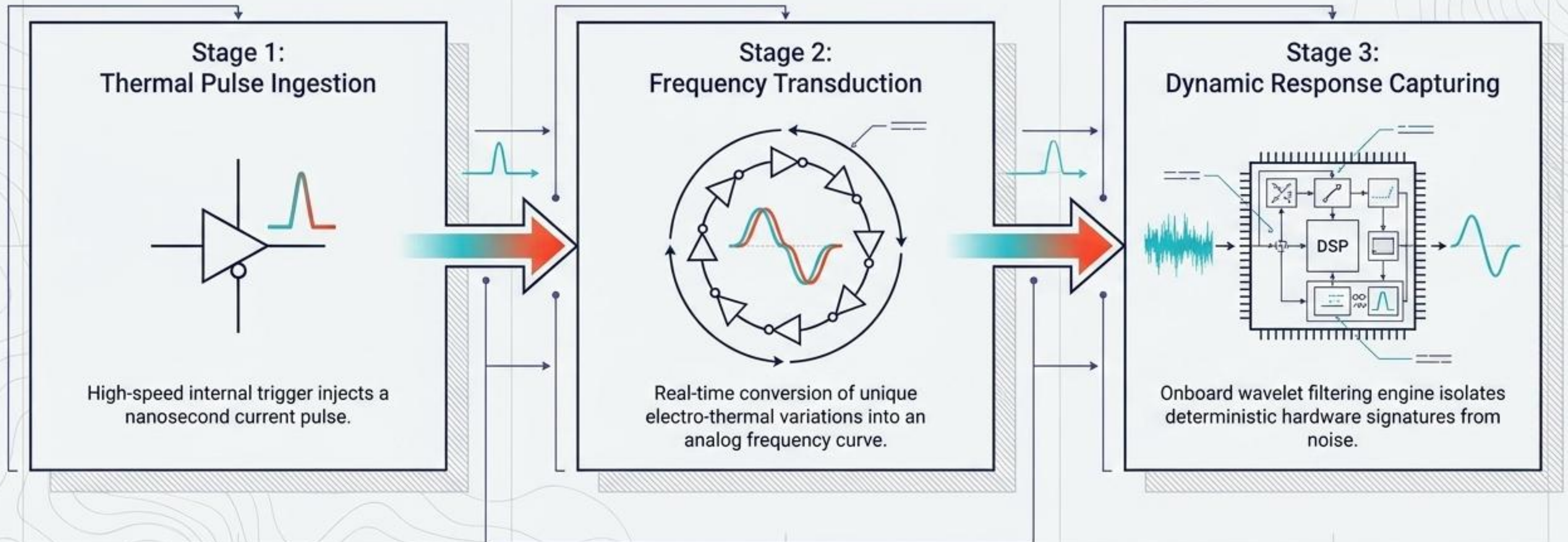


Immutable Digital Truth
(256-bit Binary)

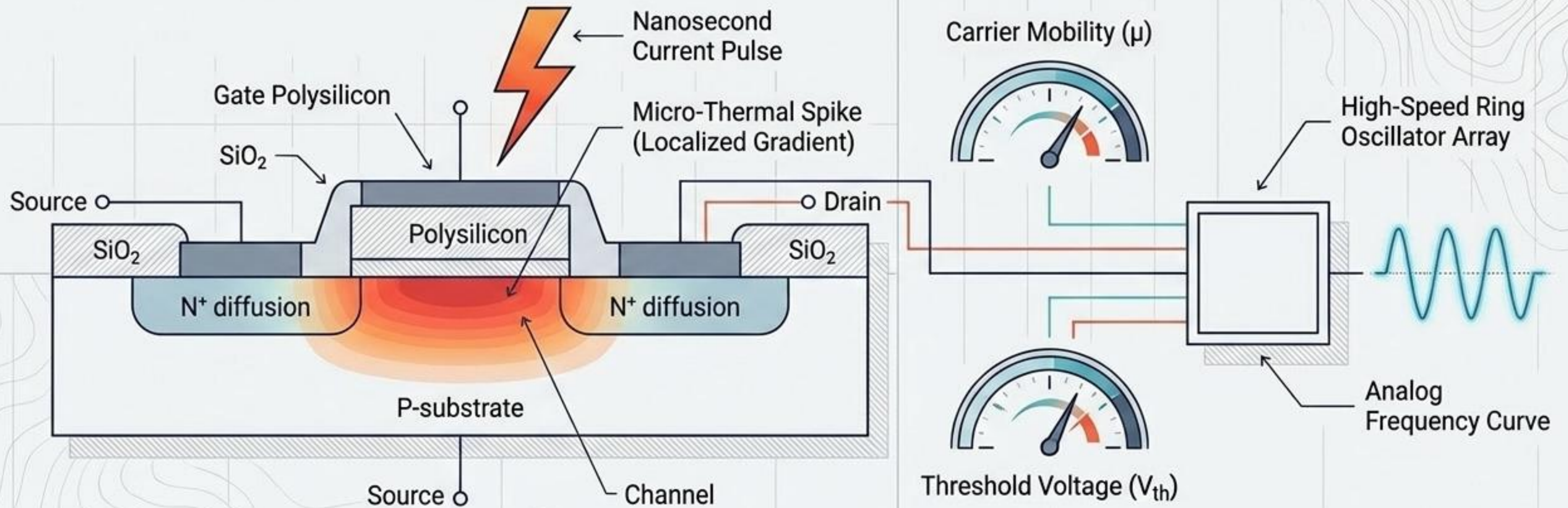


CryptoSilicon captures unpredictable atomic variance through a thermal pulse, distills it via multi-resolution math, and outputs an uncloneable hardware signature immune to aging.

A Post-Silicon Authentication Architecture



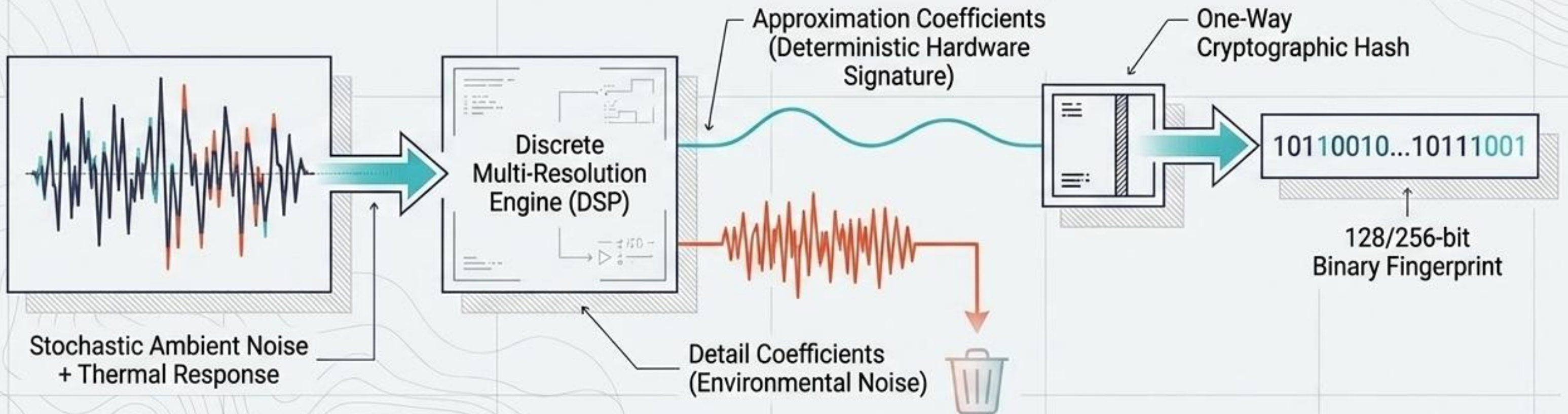
The Thermal-Transient Engine



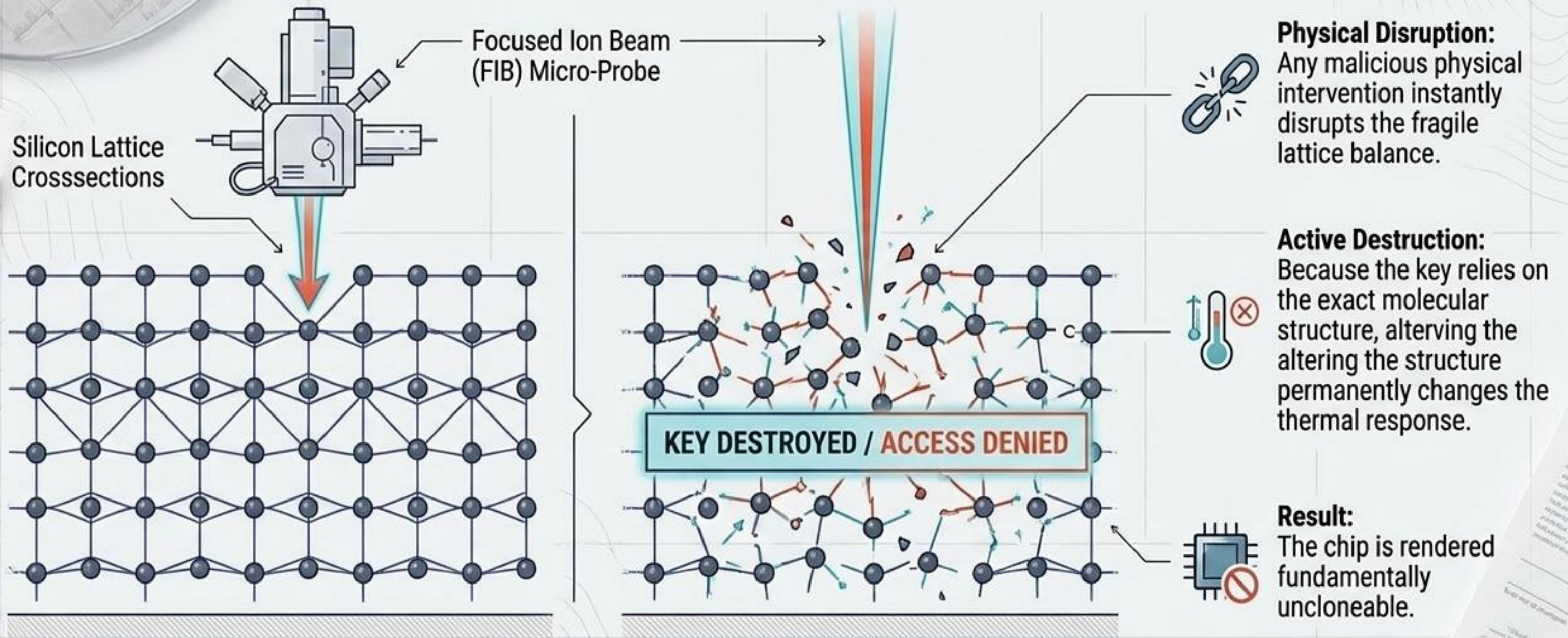
The instantaneous temperature escalation interacts with localized atomic variations, causing the rate of resistance change and thermal dissipation to vary uniquely per chip.

The Wavelet Decomposer: Distilling Signal from Chaos

$$W(a, b) = \int x(t) \psi\left(\frac{t-b}{a}\right) dt$$



Absolute Tamper-Evidence at the Molecular Level



The Definitive Security Matrix

	Standard SRAM PUF	Software Cryptography	CryptoSilicon (Thermal-Transient)
Uncloneability	○ Medium - Clonable via FIB	○ Zero - Easily copied	● Absolute - Bound to molecular structure
Tamper-Evidence	○ Low - Passive scanning possible	○ Zero - Firmware dumping	● High - Physical probing alters structure
Noise Resilience	○ Low - Prone to aging & bit-flips	● High - Mathematical	● High - Filtered via onboard DWT
Hardware Overhead	○ Medium - Requires dedicated array	○ High - Heavy crypto cores	● Ultra-Low - Standard CMOS

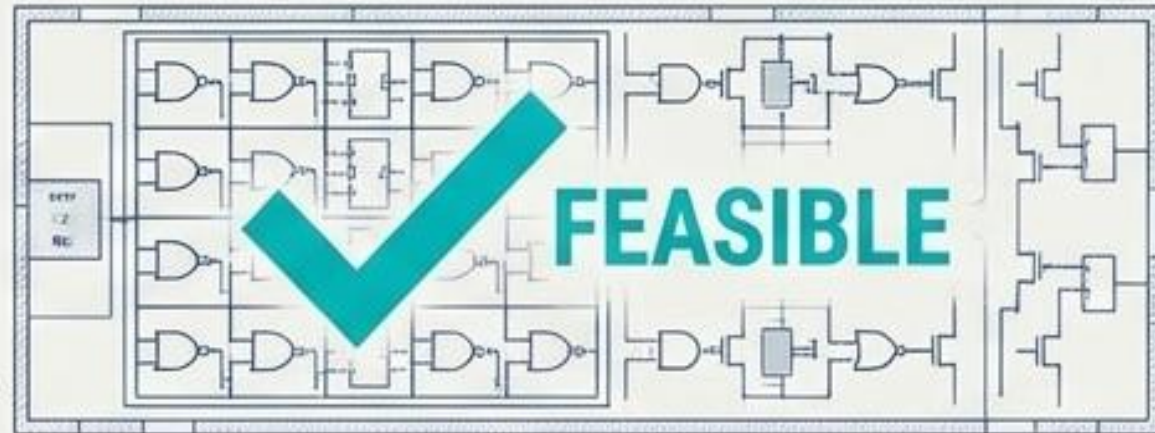
Ultra-Low Overhead & Immediate Commercial Feasibility

Architectural Contrast Diagram

Custom
Lithography
Fab



Standard
Commercial
CMOS Layout



Zero Custom Lithography Cost:
Implemented entirely using standard
commercial CMOS building blocks.
No exotic fabrication required.

Standard Silicon Aging: Resilient
to aging and bit-flips; the electro-
thermal response remains embedded
in bulk physical properties.

Target Market Sectors



High-end Automotive



Secure Cloud Servers



Defense Supply Logistics



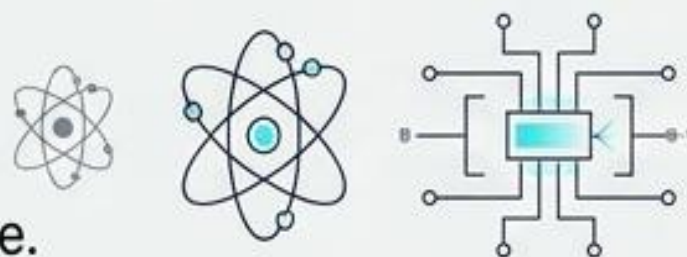
Advanced IoT Nodes

Academic Grounding & Scientific Validation

Pillar 1: The Physics

Maes, R. (2013). *Physically Unclonable Functions*. Springer.

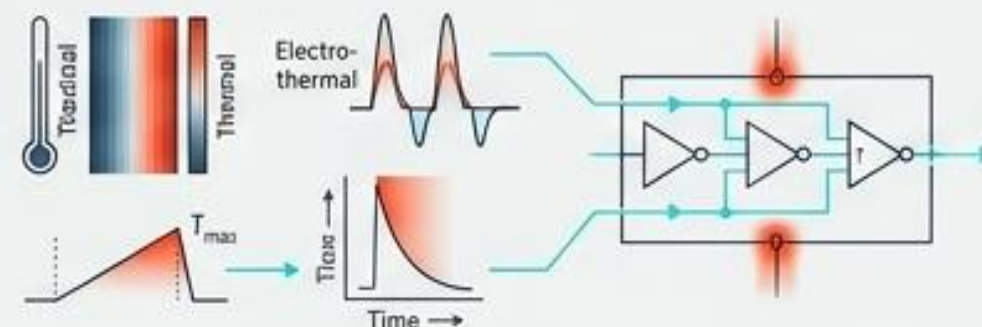
Establishes the fundamental laws of hardware fingerprinting and atomic variance.



Pillar 2: The Transduction Engine

Bhargava & Mai (2013). *IEEE Transactions on CAD*.

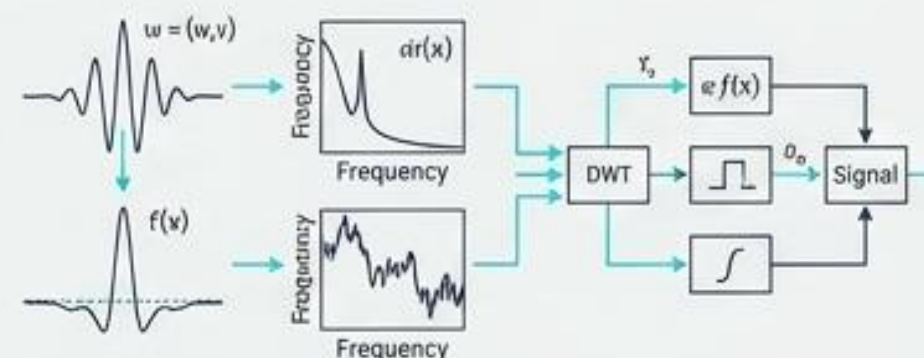
Validates the specific electro-thermal pulse and Ring Oscillator model for high-reliability authentication.



Pillar 3: The Mathematics

Mallat, S. (1999). *A Wavelet Tour of Signal Processing*. Academic Press.

Proves the efficacy of Discrete Wavelet Transforms (DWT) for stripping stochastic channel noise from analog signals.



The Molecular Standard for Global IC Security

Transitioning silicon security from hackable soft-code configurations into uncloneable atomic arrangements.

Securing global technological infrastructure, one microchip at a time.